



Human–Computer Interaction Case Study

DineWise : Behavior-Based Personalization for Food Delivery Applications

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Abstract

This project introduces a smart food delivery application called *DineWise*, which aims to improve user experience through personalized recommendations based on user behavior. The system analyzes users' ordering patterns, including preferred meals, time of ordering, and frequently selected restaurants, to generate context-aware suggestions.

Unlike traditional food delivery applications that rely on generic recommendations and advertisements, the proposed system focuses on behavioral-based personalization to reduce decision-making time and improve efficiency. The application also integrates manual filtering options, allowing users to customize their experience based on preferences such as price, delivery time, and food category.

The main goal of this project is to minimize decision fatigue, enhance usability, and provide a more adaptive and intelligent food ordering experience.

Keywords: Food Delivery, Personalization, User Behavior, Recommendation Systems, User Experience, Decision Fatigue

Introduction

Food delivery applications have become an essential part of modern daily life, especially for university students and busy individuals who depend on quick and convenient meal options. However, despite their popularity, most existing applications suffer from a major limitation: they provide generic recommendations that do not reflect individual user preferences or habits.

This issue leads to several problems, including long browsing times, difficulty in decision-making, and user frustration. Many users spend a significant amount of time searching, comparing, and deciding what to order, particularly when they are in a hurry or unsure of their choices. Additionally, current systems fail to recognize recurring user behaviors, such as ordering specific meals on certain days or times.

To address this problem, this project proposes a smart food delivery application that uses behavioral analysis to generate personalized recommendations. The system learns from user habits and automatically suggests relevant options at the right time, helping users make faster and more accurate decisions.

The application can be used in various contexts, including daily meal ordering for individuals, family meal planning, and supporting restaurant managers in understanding demand patterns.

However, the system has some limitations. It relies on collecting user data over time to build accurate recommendations, which may raise privacy concerns. Additionally, the effectiveness of the system depends on the availability of sufficient user interaction data, especially for new users.

Background

Current food delivery platforms, such as HungerStation and The Chefz, have revolutionized the way users access meals by providing a wide variety of restaurant options, secure payment gateways, and real-time tracking features. These platforms excel in providing a user-friendly interface that simplifies the basic ordering process for millions of users.

Strengths and Weaknesses of Existing Systems

While existing applications are highly functional, our analysis identifies several areas where they fall short of meeting evolving user needs, particularly regarding personalization.

Feature	Strengths	Weaknesses
Recommendation Engines	Effective for discovering new promotions and sponsored restaurants.	Rely on generic data; fail to account for individual daily routines or specific habits.
User Interface	Provides fast ordering and reliable real-time delivery tracking.	Overwhelms users with too many options, leading to significant decision fatigue.
Search & Discovery	Offers a wide range of filters (e.g., categories, price, ratings).	Lacks smart filtration, often requiring 15–30 minutes of manual browsing.
Personalization	Displays popular items and historical order lists.	Fails to recognize temporal patterns (e.g., ordering coffee in the afternoon vs. family meals on weekends).

The Gap in Current Solutions

The primary weakness in the current market is the Paradox of Choice. Users often spend an excessive amount of time scrolling past irrelevant advertisements and generic suggestions before finding a meal that fits their current context. This search friction is especially prominent for users who do not have a specific restaurant in mind and rely on the app for inspiration.

DineWise addresses these weaknesses by introducing a layer of behavioral-based technology. Unlike existing systems that treat every user the same regardless of the time or day, our proposed solution bridges the gap between convenience and true personalization by adapting the interface to match the user's unique lifestyle and recurring habits.

User Interface

Implementation Process

The implementation of the DineWise interface focused on transforming user needs into a clear and efficient design.

Initial insights were gathered from user interviews and analysis, which highlighted key issues such as decision fatigue, excessive browsing time, and lack of meaningful personalization in existing food delivery applications. These findings guided the overall design direction without directly shaping the interface structure.

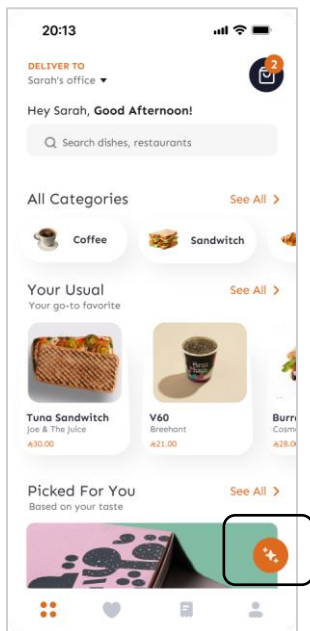
Based on these insights, we designed a user interface that emphasizes simplicity, personalization, and fast interaction. The design process began with defining the main layout and organizing key elements such as the home screen, recommendation sections, and navigation flow.

We then developed the interface using **Figma** [4], creating a high-fidelity prototype that represents the final system. During this stage, we focused on:

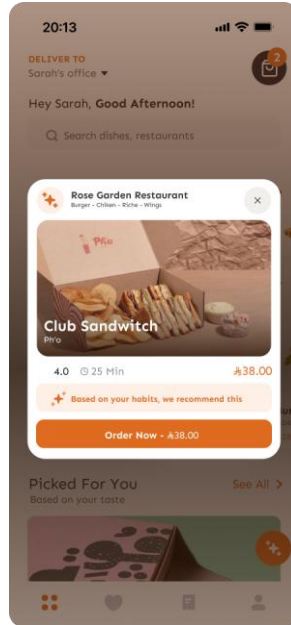
- Clear and structured layout
- Easy navigation through tap-based interactions
- Highlighting personalized features such as “Your Usual” and smart suggestions
- Designing time-based variations of the home screen (morning and night modes)

Finally, the design was iteratively refined to improve clarity, visual consistency, and usability, ensuring that users can quickly interact with the system and make decisions with minimal effort.

User Interface Screens (Input & Output)



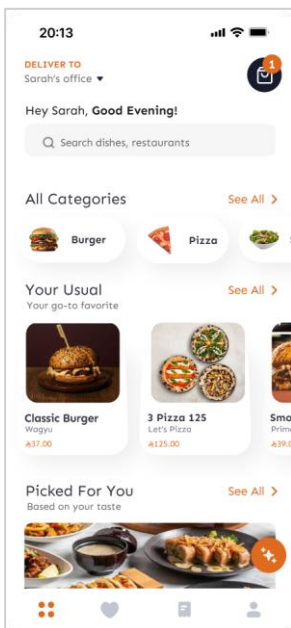
Input(1): User taps the orange suggestion button on the home screen.



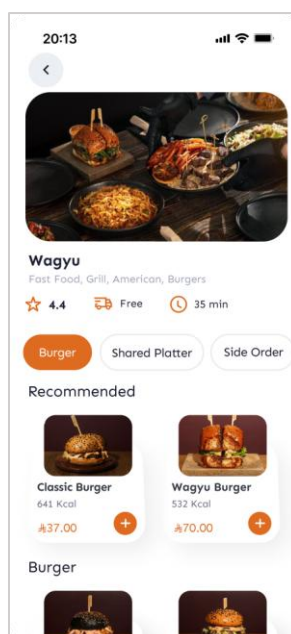
Output(1): A suggestion popup appears with recommended food and an "Order Now" option.



Input(2): User opens the DineWise application.
Output(2): The splash screen appears, displaying the DineWise logo and



Input(3): User taps on a restaurant from the home screen.



Output(3): The restaurant page is displayed, showing menu items, and pricing.

UX Pilot Evaluation

- **Visual Clarity:** Fails text contrast ratio. The light gray text falls below the WCAG 2.1 [1] AA accessibility standards for readability.
- **Home Screen:** Demonstrates strong structural layout and grid adherence (spacing and margins are consistent).
- **Navigation:** Missing navigation accessibility. Warning flagged for relying solely on iconography without descriptive text labels.
- **Consistency:** Utilizes familiar UX patterns and standard e-commerce component libraries successfully.
- **Feedback:** Standard system feedback elements are present and functionally placed.
- **Interactivity:** Touch targets are under 44px. The add buttons fall below standard mobile OS guidelines for clickable area sizes [2].

Comparison Table

Evaluation Area	My Evaluation	UX Pilot Evaluation
Visual Clarity	Clean layout, low contrast	Fails text contrast ratio
Home Screen	Smart AM/PM contextual menu	Strong structural layout
Navigation	Needs text labels on icons	Missing nav accessibility
Consistency	Consistent brand colors	Familiar UX patterns
Feedback	Clear active states	Standard system feedback
Interactivity	Add buttons are too small	Touch targets under 44px

Summary and Conclusion

Summary: Comparing the two evaluations highlights the different strengths of human vs. AI analysis. The UX Pilot AI was excellent at providing strict technical validation, it translated my general feeling that the text was "hard to read" and buttons were "too small" into exact rule violations (WCAG contrast failures and touch targets under 44px). However, the AI did not recognize the usability value of the AM/PM contextual feature, whereas my human evaluation recognized how much value that adds to the user experience.

Conclusion: This comparison showed me that AI tools are highly effective for catching technical accessibility and structural errors, validating design choices with hard data. However, human evaluation is still essential for understanding context, empathy, and the emotional flow of an app. Using both methods together helped validate the structural integrity of my design while giving me exact, actionable steps (like increasing the "+" button size and darkening the gray text) to improve its overall usability.

Testing Results

The usability testing results demonstrate that the DineWise interface is clear, efficient, and easy to use. A total of six participants were asked to complete a set of common tasks such as browsing, selecting a restaurant, adding items to the cart, and placing an order.

The results showed that most participants were able to complete the tasks successfully, with success rates ranging from 70% to 100%, which exceeds the acceptable usability threshold. This indicates that the interface supports smooth navigation and clear interaction flow.

In terms of usability, users found the application intuitive, especially features like the personalized home screen and the “Your Usual” section, which helped reduce decision time and effort.

However, minor issues were identified, including:

- Small button sizes affecting interaction accuracy
- Lack of text labels on navigation icons
- Low text contrast reducing readability

Despite these issues, users were still able to complete tasks efficiently, which confirms that the overall interface is usable and understandable.

Conclusion

Solution

DineWise provides a sophisticated solution to the common "paradox of choice" found in modern food delivery applications by implementing a behavior-based, intelligent recommendation system. By moving away from generic, advertisement-driven suggestions and toward personalized automation, the application successfully reduces decision fatigue and streamlines the ordering process. Features such as the AM/PM contextual home screen and the "Your Usual" quick-reorder shortcut allow users to transition from browsing to ordering in a matter of seconds, directly addressing the search friction identified in current market leaders.

Effectiveness

The effectiveness of this solution was validated through comprehensive usability testing across six participants, where most users achieved a task completion rate between 80% and 100%, significantly exceeding the 70% usability threshold. Users specifically highlighted the "Your Usual" section and the smart contextual menus as features that added immense value to their daily routines. Furthermore, AI-assisted evaluations confirmed that the design utilizes familiar UX patterns and maintains strong brand consistency, while also identifying technical accessibility issues in alignment with established standards such as WCAG 2.1 [1] and platform guidelines like Apple's Human Interface Guidelines [2]. Additionally, the use of AI tools such as UX Pilot [3] supported the evaluation process, and the prototype design was effectively developed using Figma [4].

Future Work

To further evolve DineWise and enhance its market competitiveness, future work should prioritize immediate accessibility improvements, including fixing text contrast ratios and increasing touch target sizes for buttons to prevent misclicks, in accordance with accessibility standards [1][2]. Additionally, adding descriptive text labels beneath navigation icons will help eliminate user confusion observed during testing.

Beyond these interface enhancements, future iterations can focus on integrating more advanced AI algorithms to improve recommendation accuracy, implementing stronger privacy controls to address user concerns, and expanding the system to support voice assistants or wearable devices. Furthermore, conducting usability testing with a larger and more diverse group of users will help refine the system and ensure scalability, supported by continuous design and prototyping tools such as Figma and AI-driven evaluation platforms like UX Pilot.

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